

E-Commerce Customer & Sales Analysis

Short Project Report

1. Project Objective

The objective of this project is to analyse the UCI Online Retail dataset using Python. The analysis covers data cleaning, exploratory sales analysis, visualisation, RFM customer analysis, and K-Means customer segmentation. The final goal is to identify meaningful customer and sales patterns and convert them into practical business recommendations.

2. Dataset

The UCI Online Retail dataset contains more than 500,000 transaction records from a UK-based online retailer. The fields include invoice number, product code, product description, quantity, invoice date, unit price, customer ID, and country.

3. Data Cleaning

The dataset was cleaned by removing duplicate rows, cancelled invoices, non-positive quantities, and non-positive prices. Missing customer IDs were retained for overall sales analysis but excluded from customer-level RFM analysis, where a valid customer identifier is required.

4. Feature Engineering

New features were created for Revenue (Quantity x UnitPrice), Month, Day of Week, and Hour. These variables support trend analysis and provide a clearer view of when and where revenue is generated.

5. Key Results

Metric	Result
Total Revenue	£10,642,110.80
Unique Orders	19,960
Identifiable Customers	4,338
Top Revenue Market	United Kingdom
Peak Full Month	November 2011

6. Customer Segmentation

Segment	Average Behaviour	Customers
High-Value Loyal	Very recent, frequent, high spending	713
At-Risk / Inactive	Long recency, low frequency, low spending	1,622
Recent / Promising	Recent purchases, lower repeat frequency	837
Regular Customers	Moderate recency, frequency and spend	1,166

The RFM variables were log-transformed and standardised before K-Means clustering. Four clusters were selected as a practical balance between statistical separation and business interpretability.

7. Main Business Findings

- The United Kingdom is the dominant revenue market, with the strongest international markets including the Netherlands, EIRE, Germany, and France.
- Revenue strengthens significantly in the later months of 2011, with November showing the highest full-month revenue.
- A relatively small high-value segment contributes much higher average spend and purchase frequency than other segments.
- The largest customer segment is At-Risk / Inactive, indicating a substantial re-engagement opportunity.

8. Business Recommendations

1. Retain high-value loyal customers through loyalty programmes, VIP treatment, and personalised offers.
2. Re-engage inactive customers using targeted email campaigns, time-limited discounts, and relevant recommendations.
3. Encourage recent customers to make repeat purchases through follow-up promotions and product suggestions.
4. Maintain adequate stock for top-selling products to minimise lost sales during periods of strong demand.
5. Protect the UK core market while developing strong international markets.
6. Prepare inventory and marketing activity before peak seasonal demand.
7. Repeat RFM segmentation periodically so marketing actions reflect changing customer behaviour.

9. Conclusion

The project demonstrates an end-to-end data science workflow using Python, Pandas, Matplotlib, Scikit-learn, feature engineering, RFM analysis, and K-Means clustering. The analysis produces actionable insights for sales planning, customer retention, re-engagement, and market development.